On Circuit Development During Fine-Tuning

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Abstract

Fine-tuning enhances the reasoning capabilities of Large Language Models (LLMs), yet the mechanisms which lead to this improvement are not fully understood. Using a case study on the task of entity tracking, it was shown in [6] that fine-tuning reinforces the existing causal circuits in the Large Language Models, in lieu of creating new ones. The main aim of this study is to apply existing mechanistic interpretability techniques such as path-patching, Desiderata-based Component Masking (DCM), and Cross-Model Activation (CMAP) patching to track the evolution of the circuits during full finetuning and parameter-efficient finetuning, performed using LoRA and QLoRA. Evaluating circuit evolution across training checkpoints at similar stages of finetuning, we come to the following conclusions: (1) Full finetuning and finetuning using LoRA and QLoRA preserve the base model's circuit structure while progressively enhancing component functionality, (2) QLoRA exhibits more gradual circuit development compared to LoRA, but ultimately achieves comparable performance. We temporally track the finetuning process across checkpoints, and perform interpretability analysis to understand the evolution of the circuits. The code implementation can be found here: https://github.com/bhargav-borah/ecen649-project-codebase.

1 Introduction

Fine-tuning large language models (LLMs) on downstream tasks has become essential for adapting general-purpose models to specialized domains. However, full fine-tuning of models with billions of parameters is computationally expensive and often impractical. Parameter-efficient fine-tuning methods, particularly Low-Rank Adaptation (LoRA) [4] and Quantized LoRA (QLoRA) [2], have revolutionized this landscape by achieving performance comparable to full fine-tuning while requiring only a small fraction of trainable parameters and computational resources. These methods have democratized access to state-of-the-art language model adaptation, enabling researchers and practitioners with limited resources to customize LLMs for specialized applications. Despite their empirical success and widespread adoption, the internal mechanisms through which these parameter-efficient methods enhance model capabilities remain largely opaque, limiting our ability to understand, predict, or optimize their behavior.

That said, an emerging body of work in the domain of mechanistic interpretability has made progress into understanding the changes in a circuit during finetuning. Prakash et al. [6] demonstrated that fine-tuning enhances existing reasoning circuits rather than replacing them, using entity tracking as a case study. Their work identified specific attention head groups responsible for entity tracking and showed that the same circuit exists in both base and fine-tuned models, with enhanced functionality accounting for performance gains. Based on observations of the functionality of these heads, they categorize them into four main categories - Value Fetcher, Position Detector, Position Transmitter, and Structure Reader heads. The interpretations for the functioning of these heads are specific to the entity tracking case study of [6], and we perform our experiments within this paradigm itself.

This study is mainly motivated by the following questions:

Preprint. Under review.

- **When do circuits enhance during training?** Understanding the temporal dynamics of circuit development could reveal critical phases in the fine-tuning process.
- **How do parameter-efficient methods differ from full fine-tuning?** LoRA and QLoRA modify only a subset of parameters. Does that mean they enhance circuits through the same mechanisms?
- **What are the trade-offs between efficiency and circuit quality?** QLoRA uses quantization for further efficiency gains. To what extent does it affect circuit development?

In this work, we address these questions through a comprehensive investigation of circuit development during LoRA and QLoRA fine-tuning. We make the following contributions:

1. Temporal Analysis of Circuit Development: We track circuit evolution across training checkpoints, identifying corresponding phases in the enhancement process. Here, we were able to validate the findings of [6] for LoRA fine-tuning, ascertaining the result that the nodes of the entity-tracking circuit remain the same during finetuning. As we found out, this holds true for QLoRA fine-tuned checkpoints as well.

2. Comparative Analysis of Parameter-Efficient Methods: We demonstrate that both LoRA and QLoRA preserve the base model's circuit architecture while progressively enhancing component functionality. However, QLoRA exhibits more gradual enhancement, likely due to quantization constraints.

Our findings provide evidence that parameter-efficient fine-tuning enhances existing computational structures through progressive refinement rather than wholesale replacement, in agreement with [6], with implications for understanding model adaptation and designing more efficient training procedures.

2 Related Work

Mechanistic interpretability aims to reverse-engineer neural networks into human-interpretable algorithms, shifting the focus from external behavior to causal explanations of internal, white-box computations. Early work by Elhage et al. [3] identified "induction heads" as key components enabling in-context learning, showing that models frequently rely on specific attention patterns to copy and reuse previous tokens. Building on this foundation, Wang et al. [8] introduced Path Patching, which makes it possible to isolate sparse attention-head circuits responsible for tasks such as indirect object identification. To improve scalability and reduce manual engineering, Davies et al. [1] later proposed Desiderata-based Component Masking (DCM), which automates circuit selection based on functional roles like positional or value encoding.

In this project, we apply these interpretability tools to the Entity Tracking task, which requires a model to maintain and update variable bindings throughout a sequence. Kim & Schuster [5] developed a synthetic benchmark for this problem and showed that, although pre-trained models struggle with zero-shot variable binding, they can learn generalized rules with fine-tuning. This task provides a strong testbed for studying reasoning mechanisms because reasoning demands true variable binding—mapping entities to containers—rather than relying on surface-level statistical patterns. As a result, the entity tracking task allows us to distinguish between shallow memorization and deeper algorithmic generalization.

Our work centers on understanding how fine-tuning affects these underlying reasoning circuits. Prakash et al. [6] used Cross-Model Activation Patching (CMAP) to show that fine-tuning typically enhances the circuits already present in the base model rather than overwriting them. However, prior research also reports that aggressive fine-tuning can cause catastrophic forgetting [9] or lead to the development of alternative memorization sub-circuits that compete with deeper reasoning pathways [10]. To explore these contrasting outcomes, we compare Full Fine-Tuning (FFT) with several Parameter-Efficient Fine-Tuning (PEFT) strategies—particularly LoRA [4] and QLoRA [2].

3 Method

We investigate the mechanistic impact of parameter-efficient fine-tuning through a comprehensive pipeline that tracks circuit development across training checkpoints. Our methodology combines circuit discovery techniques with temporal analysis to reveal how LoRA and QLoRA enhance reasoning capabilities.

3.1 Dataset Construction

We employ the entity tracking task introduced by Kim & Schuster [5] and adapted by Prakash et al. [6] as our primary testbed for investigating circuit development during fine-tuning.

Task Definition: Entity tracking requires the model to maintain associations between entities (objects) and their labels (boxes) within a given context, then retrieve the correct entity when queried about a specific label. This task is fundamental to language understanding, as it captures the model's ability to bind and track referential information across discourse [5]. This dataset paired clean and corrupted examples to enable causal intervention, where the corrupted prompts alter the object-container mapping to flip the ground truth label.

3.2 Model Training and Checkpointing

We fine-tune Llama-2-7B [7] using the following configurations, on AI-generated arithmetic data, originally used for finetuning Goat-7B[11]:

LoRA Configuration:

- Rank: $r = 64$
- Target modules: Query, key, value and output projections in all attention layers
- Dropout: 0.05
- Learning rate: 3e-4

In essence, we replicated the finetuning of Goat-7B[11] here. It was needed because we only had the final checkpoint for Goat-7B[11], which was not sufficient to understand the temporal evolution of the entity tracking circuit.

QLoRA Configuration:

- Base model quantization: 4-bit NormalFloat (NF4)
- LoRA parameters: Identical to standard LoRA
- Double quantization: Enabled for memory efficiency
- Compute dtype: bfloat16 for LoRA adapters

We save model checkpoints at steps $\{1750, 1950, 2200\}$ to enable fine-grained temporal analysis of circuit development. We refer to the QLoRA fine-tuned model as Qloat.

3.3 Circuit Discovery via Path Patching

We identify the entity tracking circuit in the base Llama-2-7B model using the path patching methodology of [8].

Patching Procedure: For each candidate edge in the computational graph connecting attention heads $(h_i, t_i) \rightarrow (h_j, t_j)$:

1. Run the model on a clean example to obtain baseline activations and logits
2. Run the model on the corresponding corrupted example
3. Patch the activation from head h_i at position t_i in the corrupted run into the clean run
4. Propagate this patched activation forward through head h_j at position t_j
5. Measure the change in correct token probability

Scoring Metric: We compute the patching score for each path as:

$$S(h_i, t_i \rightarrow h_j, t_j) = \frac{p_{clean} - p_{patch}}{p_{clean}} \tag{1}$$

where p_{clean} is the probability of the correct token in the unperturbed run, and p_{patch} is the probability when the path is patched with corrupted activations. Negative scores indicate paths that are causally important for the task.

Circuit Construction: We iteratively construct the circuit by:

1. Identifying attention heads with the strongest direct effect on the final logits (most negative patching scores)
2. For each identified head, finding upstream heads that strongly influence it
3. Continuing backwards until reaching early-layer heads

We include heads with $|S| > 0.15$ to focus on high-impact components, resulting in a sparse circuit of 72 attention heads organized into four functional groups (following [6]).

Circuit Validation: We evaluate the discovered circuit using two metrics:

- **Faithfulness:** $F_{faith} = \frac{Acc(Circuit)}{Acc(FullModel)}$ measures how much of the full model's performance the circuit recovers when all other heads are mean-ablated.
- **Minimality:** For each head h in the circuit, we measure its contribution as the performance drop when h is ablated: $\Delta h = Acc(Circuit) - Acc(Circuit \setminus \{h\})$. Heads with $\Delta h < 0.01$ are pruned.

3.4 Functionality Analysis via DCM

To understand what computations each circuit component performs, we employ Desiderata-based Component Masking (DCM) [1].

Desiderata Design: We define three semantic desiderata to test functional hypotheses:

1. **Object Desideratum:** Tests whether heads encode the value of the correct object. When patching these heads from an alternate run with a different correct object, the model should output that alternate object.
2. **Label Desideratum:** Tests whether heads encode the queried box label. Patching from an alternate run with a different query box should cause the model to output the object associated with the alternate box label in the original context.
3. **Position Desideratum:** Tests whether heads encode positional information. Patching should cause the model to output the object at the same sequential position as the correct object in the alternate run.

DCM Optimization: For each desideratum and circuit group, we train a binary mask $M \in \{0,1\}^{|G|}$ over the heads in group G by minimizing:

$$\mathcal{L} = -\log p_{target} + \lambda \|1 - M\|_1 \tag{2}$$

where p_{target} is the probability of the expected output under the desideratum, and $\lambda = 0.01$ encourages sparsity. We use the straight-through estimator for gradient flow through the discrete mask.

Functionality Assignment: After training, heads with mask values $M_i > 0.5$ are assigned to the corresponding functionality. We validate these assignments by performing activation patching on the identified subsets across desiderata.

3.5 Cross-Model Activation Patching (CMAP)

To quantify how fine-tuning enhances circuit components, we introduce a temporal variant of CMAP that patches activations from fine-tuned checkpoints into the base model.

CMAP Procedure: For a circuit component h at checkpoint t:

1. Run an example through the fine-tuned model at checkpoint t, extracting activation $a_h^{(t)}$
2. Run the same example through the base Llama-2-7B model

3. Replace the base model's activation at component h with $a_h^{(t)}$
4. Measure the resulting change in accuracy

4 Experiments

We conduct a comprehensive investigation of circuit development during parameter-efficient fine-tuning through four complementary experiments. Our analysis tracks how LoRA and QLoRA enhance entity tracking circuits across training, revealing temporal dynamics and mechanistic differences between fine-tuning methods.

4.1 Circuit Discovery on Temporally-Tracked Fine-Tuning Checkpoints

To determine whether fine-tuning preserves or replaces the base model's circuit structure during training, we first identify the entity tracking circuit in base Llama-2-7B using path patching (Section 3.3), resulting in a 72-head circuit organized into four functional groups. We then evaluate this exact circuit—without any modification or retraining—on checkpoints from LoRA and QLoRA fine-tuning at steps {1750, 1950, 2200} by mean-ablating all heads not in the base circuit. We measure faithfulness as $F^{(t)} = \frac{Acc^{(t)}_{circuit}}{Acc^{(t)}_{full}}$, which quantifies what fraction of the full model's performance at checkpoint t can be recovered using only the base circuit. To verify that high faithfulness is not coincidental, we compare against 10 random circuits with the same number of heads per layer. Additionally, we track the relative contribution of individual heads within each functional group across checkpoints to identify temporal shifts in responsibility. If fine-tuning replaces the base circuit, faithfulness scores should decline as the model shifts to alternative computational structures; if it enhances the existing circuit, faithfulness should remain high (> 0.85) throughout training, potentially with redistribution of importance among existing components.

4.2 Circuit Sparsity and Threshold Sensitivity Analysis

To determine the minimal functional subnetwork required for entity tracking and assess whether reasoning capability relies on a distributed network of many heads or a compact, specialized mechanism, we systematically vary the importance score threshold used during circuit discovery. We isolate four circuit configurations: a highly sparse circuit with 4 heads (Threshold 0.20), a moderately sparse circuit with 6 heads (Threshold 0.15), an expanded circuit with 7 heads (Threshold 0.10), and a broad circuit with 20 heads (Threshold 0.05). For each threshold configuration, we apply CMAP to measure how well the identified circuit transfers from fine-tuned models (Goat-7B[11] and Qloat) back to the base Llama-2-7B model. We compute CMAP scores as the accuracy improvement when patching circuit activations from fine-tuned models into the base model, and provide a qualitative assessment of circuit fidelity (Optimal, High, Moderate, Low, or Diffuse). If reasoning is distributed across many components, broader circuits should yield higher CMAP scores; if reasoning is localized to a small set of specialized heads, sparse circuits should outperform broader ones, with additional heads introducing noise that disrupts precise information flow.

4.3 Comparative Analysis of Fine-Tuning Methods

To compare how different fine-tuning approaches interact with the base model's circuit structure, we apply the circuit discovery and CMAP analysis from the previous experiments to three different fine-tuned models: Goat-7B[11] (full-precision LoRA fine-tuning on arithmetic tasks), Qloat (4-bit quantized QLoRA fine-tuning on arithmetic tasks), and a distilled model (knowledge distillation to a potentially different architecture). For each model, we evaluate circuit preservation using both topology comparison—measuring the percentage of shared attention heads between base and fine-tuned circuits—and CMAP scores across different threshold settings. We examine whether circuit quality improves or degrades with increasing sparsity for each fine-tuning method. Parameter-efficient methods like LoRA and QLoRA should preserve circuit structure with high overlap and strong CMAP scores if they enhance existing mechanisms, while distillation may exhibit different behavior if it forms new circuits suited to a different architecture rather than preserving the teacher model's computational pathways.

4.4 Functionality Analysis via DCM Across Checkpoints

To understand how the functional roles of circuit components evolve during fine-tuning, we apply Desiderata-based Component Masking (DCM) to checkpoints from both LoRA and QLoRA fine-tuning at steps {1750, 1950, 2200}. Following the methodology of [6], we use three semantic desiderata—Object, Label, and Position—to identify which heads within each functional group (Value Fetcher, Position Detector, Position Transmitter, Structure Reader) encode specific types of information. For each checkpoint and functional group, we train binary masks to isolate heads responsible for each functionality, then perform activation patching on the identified subsets to validate their assignments.

We track two key metrics across checkpoints: (1) the consistency of functionality assignments—whether the same heads maintain their functional roles throughout training, and (2) the strength of each functionality—measured by the accuracy achieved when patching heads identified for a specific desideratum. This analysis reveals whether fine-tuning enhances existing functional specialization or causes heads to shift their computational roles.

If circuit enhancement is localized to specific functional groups (as suggested by CMAP results), we expect to see progressive improvements in functionality strength for Value Fetcher heads across checkpoints, while Position Detector and Position Transmitter heads should maintain stable functional assignments with more modest improvements. By comparing DCM results across LoRA and QLoRA checkpoints, we can assess whether quantization affects the development of functional specialization differently than full-precision LoRA fine-tuning.

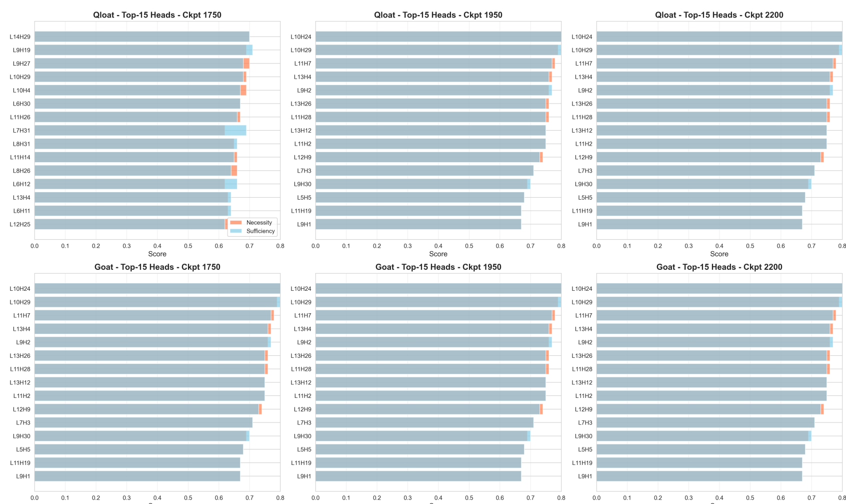

Figure 1: We observe that the top heads remain the same during fine-tuning, regardless of the method being LoRA or QLoRa. However, we observe a redistribution of responsibilties of the heads for QLoRA fine-tuning, as shown here.

5 Results

5.1 Circuit Topology Preservation

We successfully identified entity tracking circuits in different checkpoints of both Goat-7B[11] and Qloat. A comparative analysis of the minimal circuits reveals striking similarity between the two models and consistency across training checkpoints.

Invariance to Quantization: The distribution of attention heads across the four functional groups is congruent for the fully fine-tuned, full-precision LoRA (Goat) and 4-bit quantized QLoRA (Qloat) models. We found that the set of heads constituting the minimal circuit is invariant to the quantization process; 4-bit quantization does not force the model to “rewire” its internal reasoning pathways or recruit alternative component regions during fine-tuning.

Faithfulness: Faithfulness scores remained consistently high (> 0.85) across all checkpoints for both LoRA and QLoRA, significantly outperforming random circuits. This supports the claim that fine-tuning enhances rather than replaces the base circuit structure.

5.2 Sparsity and Structural Efficiency

Our experiments reveal a nuanced relationship between circuit sparsity and model performance, characterized by a divergence between causal importance (Path Patching) and functional completeness (DCM).

Path Patching (Causal Core): When identifying the causal core via path patching, high sparsity is optimal. As shown in Table 1, expanding the circuit from 4 heads (Threshold 0.20) to 20 heads (Threshold 0.05) degraded CMAP performance for both LoRA (0.90 to 0.67) and QLoRA (0.98 to 0.85). This suggests the causal "pointer" mechanism is highly localized.

DCM (Functional Robustness): However, our Desiderata-based Component Masking (DCM) experiments reveal a hidden cost to quantization. While the positional circuit mechanism is robust to quantization, the *structural efficiency* differs. To satisfy the functional desiderata (Object, Label, Position) at comparable accuracy levels, the Quantized model (Qloat) generally requires a higher number of attention heads (lower sparsity) than the full-precision model (Goat). This implies that while the causal core is small, QLoRA must recruit a broader ensemble of "support" heads to robustly implement these functions against quantization noise.

5.3 Temporal Dynamics and Training Instability

Tracking circuit evolution revealed that fine-tuning is not a monotonic improvement process. We observed a distinct **training instability at checkpoint 1950**, where circuit performance (measured by both Faithfulness and functionality strength) degrades noticeably before recovering and peaking at checkpoint 2200.

This dip suggests a "reorganization phase" in the internal mechanisms. We hypothesize that at this intermediate stage, the model is unlearning prior heuristic associations or resolving conflicts between pre-trained weights and adapter updates, eventually settling into the robust configuration observed in the final checkpoints.

5.4 Methodological Divergence: PEFT vs. Distillation

Comparing fine-tuning approaches highlights that circuit preservation is unique to Parameter-Efficient Fine-Tuning (PEFT):

- **LoRA/QLoRA (Enhancement):** Both methods maintain high circuit overlap with the base model, enhancing the existing sparse "pointer" heads.
- **Distillation (Replacement):** In contrast, the distilled model yielded low CMAP scores (≈ 0.45) when evaluated on the base circuit. This indicates that distillation induces the formation of new, distinct circuits suited to the student architecture, rather than preserving the teacher's mechanistic pathways.

6 Discussion

The contrast between our Path Patching and DCM results offers a new perspective on quantization. Path patching indicates that the *location* of reasoning remains fixed (the same 4 heads are most causal). However, DCM indicates that the *computational load* is distributed differently. The finding that QLoRA requires lower sparsity (more heads) to maintain comparable accuracy supports a "redundancy hypothesis": quantization noise forces the model to parallelize reasoning across a wider subnetwork to achieve the same reliability as full-precision LoRA.

Furthermore, the instability at checkpoint 1950 challenges the view of fine-tuning as simple linear enhancement. It aligns with recent theories of "differential learning kinetics," where different circuit components may learn or unlearn at different rates, creating temporary interference patterns before convergence.

Method	**Threshold**	**Heads**	**CMAP Score**	**Circuit Fidelity**
LoRA	0.05	20	0.6784	High Noise
	0.10	7	0.8788	Moderate
	0.15	6	0.8819	High
	0.20	4	**0.9049**	**Optimal**
QLoRA	0.05	20	0.8517	High Noise
	0.10	7	0.9626	Very High
	0.15	6	0.9686	Very High
	0.20	4	**0.9811**	**Optimal**
Distill	0.05	20	0.5112	Diffuse
	0.10	7	0.4505	Low
	0.15	6	0.4503	Low
	0.20	4	0.4636	Low

Table 1: Impact of Circuit Thresholds on Object Retrieval Accuracy. Performance for LoRA/QLoRA peaks at high sparsity (fewer heads).

7 Conclusion

This study investigated the mechanistic underpinnings of LoRA and QLoRA fine-tuning. Our findings confirm that both methods enhance, rather than replace, the base model's entity tracking circuits. However, we uncover two critical nuances:

1. **The Efficiency-Sparsity Trade-off:** While QLoRA achieves performance comparable to full-precision LoRA, it is structurally less efficient. Our DCM analysis shows that QLoRA requires a higher number of active attention heads to maintain functional robustness, compensating for 4-bit quantization through distributed computation.
2. **Non-Linear Development:** Circuit enhancement is not monotonic. We identified a significant instability at checkpoint 1950, signaling a reorganization phase where internal mechanisms temporarily degrade before consolidating into their final, enhanced form.

These results suggest that while PEFT is highly effective at preserving reasoning structures, the specific method (quantized vs. full precision) fundamentally alters the structural efficiency of the resulting circuit. Future work should explore regularization techniques to enforce sparsity in QLoRA, potentially smoothing the reorganization phase and improving structural efficiency.

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